

Explainable Machine Learning for Early Pregnancy Loss Prediction Using Longitudinal Ultrasound Data

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ABSTRACT

Early pregnancy loss remains a common obstetric complication, which affects nearly one in five clinically recognized pregnancies. The fetal heart rate and early ultrasound growth parameters serve as key indicators of embryonic viability. However, the clinical interpretation of these indicators relies largely on fixed thresholds and subjective assessment, which limits predictive reliability across diverse populations. Current prediction approaches treat fetal heart rate as an isolated, static measurement and often ignore the combined influence of ultrasound growth metrics and maternal clinical factors. Moreover, most artificial intelligence models lack transparency, which reduces clinical trust and limits practical adoption. The absence of longitudinal modeling further constrains the ability to capture early physiological deviations that precede pregnancy loss. This study proposes an explainable, data-driven framework that integrates fetal heart rate, crown-rump length, yolk sac diameter, gestational sac metrics, and maternal clinical variables within a unified predictive pipeline. The method applies gradient-boosted decision trees as the primary learner, which captures nonlinear feature interactions in structured clinical data. A time-aware feature construction strategy models week-to-week fetal heart rate progression that has improved temporal sensitivity without requiring dense sequential data. Model interpretability has been ensured through SHAP-based global and local explanations, which provide clinically intuitive risk attribution. Rigorous data leakage control, stratified patient-level splitting, and nested cross-validation have been applied to preserve statistical validity. The proposed framework achieves superior performance compared with existing methods. On the test dataset, the model reaches an accuracy of 0.92, sensitivity of 0.91, specificity of 0.91, and F1-score of 0.91 at epoch 1000. The Brier score improves to 0.07, indicating strong calibration and reliable risk estimation.

Keywords: Early pregnancy loss, fetal heart rate, ultrasound biomarkers, explainable machine learning, time-aware prediction

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1. Introduction

Early pregnancy represents a biologically sensitive phase that determines the viability of embryonic development and the eventual outcome of gestation. Miscarriage, which refers to spontaneous pregnancy loss before the completion of the first trimester, affects approximately 10–20% of clinically recognized pregnancies and remains one of the most common complications in obstetric practice [1]. Despite advances in prenatal imaging and maternal care, early pregnancy loss continues to impose substantial emotional, psychological, and clinical burdens on affected individuals and healthcare systems [2]. Accurate early risk assessment therefore remains a critical priority in modern obstetrics.

Background

Ultrasound examination during early gestation provides the primary noninvasive means for

assessing embryonic development and pregnancy viability. Among the measurable indicators, the fetal heart rate has emerged as a central physiological marker that reflects early cardiac function and embryonic health [3]. Clinical studies consistently show that abnormally low fetal heart rate values during the early weeks of gestation correlate with an increased risk of pregnancy loss [4]. As a result, fetal heart rate assessment has become a routine component of first-trimester ultrasound evaluation. Beyond fetal heart rate, several structural ultrasound parameters contribute important prognostic information. Crown-rump length serves as a standard metric for embryonic growth assessment and gestational dating, while yolk sac diameter and gestational sac dimensions reflect early placental and embryonic support systems [5]. Deviations in these parameters often signal impaired embryonic development and adverse pregnancy outcomes. Maternal factors, including age, body mass index,

prior pregnancy history, and chronic comorbidities, further influence early gestational stability and interact with fetal growth dynamics [6].

Advances in medical imaging, electronic health records, and computational methods have increased the availability of early pregnancy data across clinical settings. Data mining and machine learning techniques provide new opportunities to analyze complex, high-dimensional datasets that exceed the analytical capacity of traditional statistical approaches [7]. In obstetric research, these methods increasingly support predictive modeling, risk stratification, and clinical decision support [8]. However, their application to early pregnancy loss prediction remains limited and uneven.

Challenges in Existing Approaches

Despite the recognized prognostic value of early ultrasound markers, several challenges persist in current research and clinical practice. First, most studies rely on static threshold-based interpretations of fetal heart rate, such as fixed cut-off values below 100 beats per minute at specific gestational ages [9]. While such thresholds offer simplicity, they fail to account for physiological variability across individuals and developmental stages. Embryonic cardiac activity evolves dynamically across early gestation, and static thresholds inadequately capture this temporal complexity.

Second, a large proportion of existing studies focus on single predictors, most commonly fetal heart rate, while ignoring complementary ultrasound growth parameters and maternal clinical variables [10]. This reductionist approach limits predictive performance and overlooks interactions that jointly influence pregnancy viability. The early gestational environment reflects a tightly coupled system in which fetal growth, placental development, and maternal physiology interact continuously.

Third, limited dataset size and single-center data collection represent persistent methodological constraints. Many published studies rely on small cohorts drawn from individual hospitals or regional clinics, which restricts statistical power and limits generalizability across populations [11]. The absence of standardized, multi-center datasets further hinders reproducibility and comparative evaluation of predictive models.

Fourth, while machine learning methods demonstrate strong performance in other areas of medical prediction, their adoption in early pregnancy loss research remains sparse. When such methods appear, they often function as black-box models that provide little insight into feature contributions or decision logic [12]. This lack of explainability reduces clinical trust and impedes integration into routine obstetric workflows.

Finally, most existing analyses treat ultrasound measurements as isolated observations rather than as components of a longitudinal developmental trajectory. The neglect of time-dependent modeling

prevents detection of subtle deviations in growth or cardiac progression that may precede overt clinical signs of pregnancy loss [13].

Problem Statement

Given these challenges, current approaches to early pregnancy loss prediction exhibit limited accuracy, interpretability, and clinical utility. The continued reliance on subjective judgment and fixed cut-off values leads to inconsistent risk assessment and uncertainty in early counseling. The lack of integrated, longitudinal, and explainable predictive frameworks constrains the ability to identify high-risk pregnancies at a stage when clinical intervention and monitoring may still provide benefit [14].

Research Gap

A critical gap exists at the intersection of multimodal ultrasound assessment, longitudinal modeling, and explainable machine learning for early pregnancy loss prediction. Although fetal heart rate, crown-rump length, yolk sac diameter, and maternal factors individually demonstrate prognostic relevance, few studies integrate these variables within a unified predictive framework. Moreover, existing models rarely capture week-to-week physiological progression or provide transparent explanations that clinicians can readily interpret.

There remains no widely adopted approach that combines structured ultrasound data, maternal clinical variables, and time-aware feature representation while maintaining rigorous data leakage control and external validity. Addressing this gap requires a method that balances predictive accuracy with interpretability and aligns with real-world clinical workflows.

The primary objective of this research is to develop an explainable machine learning framework that predicts early pregnancy loss using fetal heart rate, ultrasound growth parameters, and maternal clinical data. The study aims to integrate multiple predictors into a cohesive model that reflects the biological complexity of early gestation. Secondary objectives include modeling temporal changes in fetal heart rate and growth indicators, evaluating model calibration and clinical utility, and assessing generalizability across diverse population cohorts. The framework seeks to provide transparent risk estimates that support clinician confidence and informed decision-making.

The novelty of this work lies in the combination of time-aware feature construction with explainable machine learning applied to early pregnancy ultrasound data. Unlike traditional static models, the proposed approach captures developmental trends that reflect physiological progression across early gestation. The integration of SHAP-based explainability enables both global and patient-specific interpretation, which bridges the gap between computational prediction and clinical reasoning.

Contributions

This study makes the following key contributions. First, it presents a comprehensive, explainable machine learning framework that integrates fetal heart rate trajectories, ultrasound growth parameters, and maternal clinical factors for early pregnancy loss prediction. Second, it demonstrates that time-aware modeling combined with transparent feature attribution improves predictive performance and clinical interpretability, thereby advancing the practical applicability of data-driven decision support in early obstetric care.

2. Literature Survey on AI/ML-based Prediction Models for Early Pregnancy Outcomes

Recent studies on early pregnancy outcome prediction increasingly integrate ultrasound imaging, clinical markers, and machine learning to improve diagnostic accuracy and reduce clinical uncertainty. Liu et al. [15] establish an artificial intelligence framework to extract biometric parameters from ultrasound videos and predict early pregnancy loss (EPL). The two-stage model first uses a modified U-Net network, A3F-net, to automatically measure gestational sac area (GSA), yolk sac diameter (YSD), crown rump length (CRL), and fetal heart rate (FHR). These biometric parameters then feed into an ensemble learning model for EPL prediction. The biometric extraction achieves high precision for GSA (98.64%), YSD (96.94%), and CRL (92.83%), outperforming competing models. Bland–Altman analysis reveals no systematic deviations between AI and physician measurements, and the relative error is minimal (0.060 ± 0.057). In the prediction stage, CatBoost demonstrates superior performance, reaching 98.0% precision and an AUC of 0.969 (95% CI: 0.962–0.975). This study demonstrates that AI can effectively eliminate manual measurement errors and provide decision support, particularly for clinicians with limited experience.

Liu et al. [16] focus on a different patient group, i.e., women with recurrent pregnancy loss (RPL). The study develops week-wise prediction models by combining ultrasound indices, demographic factors, and serum markers. The independent risk factors vary across gestational weeks, where age and progesterone (P) dominate in week five, and CRL becomes significant from week six onward. The model's AUC increases from 0.671 in week five to 0.872 in week seven, showing that predictive accuracy improves as the pregnancy progresses and more biometric features become available. The authors identify specific thresholds such as $hCG < 69,636.6$ mIU/ml and $mGSD < 18.3$ mm, which significantly increase pregnancy loss risk. This work contributes to the literature by demonstrating the value of dynamic, week-specific risk modeling, which reflects the evolving physiological changes in early pregnancy.

Ultrasound Radiomics and Multimodal Prediction in RPL

Radiomics and multimodal imaging approaches further advance the prediction of pregnancy complications by extracting subtle features beyond traditional measurements. Yan et al. [17] develop an ML model that uses radiomic features from multimodal transvaginal ultrasound images to assess endometrial receptivity (ER) in unexplained RPL patients. The study uses a five-step feature selection process and trains multiple classifiers on radiomic, clinical, and combined datasets. The combined model using XGBoost achieves the best performance, with an AUC of 0.871 in the training cohort and 0.844 in the testing cohort. SHAP analysis highlights the importance of endometrial shear wave elastography (SWE) features and clinical variables such as age, SAPI, and VI. This study reveals that radiomic features can capture microstructural and biomechanical changes in the endometrium that are critical for implantation success.

Yan et al. [18] extend this work by applying deep learning to routine clinical ultrasound images and tabular data. The study employs a pre-trained ResNet-50 model for image analysis and TabNet for clinical data, then integrates them into a fusion model. The ResNet-50 model shows superior performance among deep learning architectures, and TabNet outperforms traditional ML models in tabular data interpretation. The fusion model yields the highest predictive accuracy, with an AUC of 0.853, and is presented via a nomogram. This study emphasizes that combining imaging and clinical data improves predictive performance, and that the imaging model contributes more to the overall accuracy, indicating the strong predictive value of ultrasound features.

Semantic Feature-Based ML Models for Uterine Cavity Conditions

In parallel to RPL studies, semantic feature extraction from ultrasound reports and images is used to predict uterine cavity conditions that impact pregnancy outcomes. Zhang et al. [19] propose a structured semantic feature analysis of two-dimensional transvaginal ultrasound images to evaluate uterine cavity clinical environment (UCCE). The study integrates diagnostic reports with images to define semantic features such as endometrial thickness, echogenicity, polyps, and uterine morphology. Multiple ML models, including XGBoost, random forest, SVM, and LightGBM, are trained, and the best model achieves an AUC of 0.982 in the testing set. The high accuracy of the XGBoost model and the significant difference in nomoscores between normal and abnormal UCCE groups suggest that semantic ultrasound features can reliably characterize uterine cavity abnormalities. The study also uses decision curve analysis to confirm clinical benefits, demonstrating that the model supports decision-making in reproductive medicine.

Radiomics and Clinical Feature Fusion for Miscarriage Prediction

The use of radiomics combined with clinical data is increasingly recognized as a robust approach for miscarriage prediction. Murugesu et al. [20] conduct a multi-site retrospective study using longitudinal ultrasound images and clinical features to predict miscarriage in early pregnancy. The dataset includes 500 pregnancies from two London hospitals, split into training, validation, and external testing cohorts. The best performing model uses logistic regression with elastic net feature selection and combines radiomic and clinical features. This PUVPS model achieves a recall of 0.81 and an AUC of 0.91 in the training set, and maintains high performance in validation and external testing (AUC 0.83 and 0.82, respectively). The study demonstrates that combining radiomics with clinical data outperforms models using either feature set alone. Importantly, the study is externally validated, which increases its clinical credibility and lays the foundation for prospective trials.

Preconception Risk Prediction Models for RPL

Yang et al. [21] propose a preconception risk prediction model for subsequent early pregnancy loss in women with RPL. The study uses a prospective cohort design and evaluates multiple machine learning algorithms, including GLM, GBM, RF, and deep learning. The GBM model achieves the highest AUC of 0.805 in the training set, and the validation set confirms good discrimination (AUC 0.781, 95% CI 0.702–0.843). The study compares feature selection methods, finding no significant difference between LASSO-selected and logistic regression-selected variables. The decision curve analysis and calibration plots show that the model has good clinical utility and fit. This work is valuable because it provides a preconception tool, allowing clinicians to stratify risk before pregnancy, which can inform interventions and counseling.

Ultrasound Image-Based Gestational Age Estimation

Beyond pregnancy loss prediction, machine learning is also applied to gestational age estimation. Lee et al. [22] use ultrasound image analysis alone, without manual biometric measurements, to estimate gestational age. The model is trained and validated on two independent datasets and is blinded to the ground truth during validation. The best model achieves a mean absolute error of 3.0 days in the second trimester and 4.3 days in the third trimester, outperforming conventional ultrasound biometry. The study demonstrates that image-based deep learning models can compensate for size variation and provide accurate dating, especially in cases of intrauterine growth restriction. This approach reduces reliance on operator-dependent measurements and enhances gestational age assessment in clinical settings.

ML Models for Management of Miscarriage

Finally, Murugesu et al. [23] develop and validate ML models to predict the success of expectant and medical management of miscarriage. The study includes 1075 patients and evaluates multiple algorithms across feature sets. In the expectant management cohort, the best model achieves AUC scores of 0.72, 0.63, and 0.70 in training, validation, and external testing, respectively. In the medical management cohort, the AUC scores are 0.64, 0.62, and 0.71. These results indicate moderate predictive performance, which is clinically useful given the complexity of miscarriage management decisions. The study highlights the need for further improvement and suggests that combining additional biomarkers or imaging features may enhance prediction.

Table 1. Comparative Table of Related Works

Reference	Method (Dataset/Setting)	Algorithm	Metrics	Results	Advantage	Limitation
[15] Liu et al.	630 ultrasound videos (6–10 weeks)	A3F-net (modified U-Net) + Ensemble learning (Cat Boost best)	Dice, IOU, Precision, AUC	Precision: GS 98.64%, YSD 96.94%, CR 92.83%; AUC 0.969	Eliminates manual measurement error; provides decision support	Limited to early singleton pregnancies; generalization not tested
[16] Liu et al.	RPL patients; week-wise ultrasound + serum markers	Logistic regression + week-specific risk factors	AUC	AUC range: 0.671–0.872	Dynamic weekly model reflects physiological changes	Limited to RPL patients; lacks external validation
[17] Yavuz	346 RPL vs	XGBoost +	AUC	Training	Uses radiomics	Retrospective

n et al.	369 controls (multimodal TVUS)	radiomics + SHAP		AUC 0.871; Testing AUC 0.844	for microstructural evaluation; interpretability	study; feature selection may bias results
[18] Yan et al.	346 RPL vs 369 controls; ultrasound + clinical	ResNet-50 + TabNet + fusion	AUC, Brier score	Fusion AUC 0.853	Integrates imaging and clinical data; nomogram visualization	Limited interpretability in fusion model
[19] Zhang et al.	442 infertility cases (UCC E)	XGBoost, RF, SVM, LightGBM	AUC, Brier score	Testing AUC range: 0.902–0.982	Structured semantic features improve prediction	Relies on report-based semantic annotation
[20] Murugesu et al.	500 early pregnancies (multi-site)	Logistic regression + elastic net	AUC, Recall, F1	AUC 0.91 (train), 0.82 (external); Recall 0.81	External validation; radioomics + clinical fusion	Retrospective; requires high-quality imaging
[21] Yang et al.	1050 preconception RPL cohort	GBM, RF, GLM, DL	AUC, DC, calibration	Best AUC 0.805 (train); 0.7	Precise risk stratification	Requires large cohort; may not generalize

				81 (validation)		realize
[22] Lee et al.	Two independent datasets (2nd/3rd trimester)	Image-based ML model	MAE	MAE: 3.0 days (2nd tri), 4.3 days (3rd tri)	Eliminate manual measurement dependence	Not focused on pregnancy loss prediction
[23] Murugesu et al.	1075 miscarriage management cases	Logistic regression + LASSO/RF	AUC	AUC: 0.70–0.72 (expectant), 0.71 (medical)	Supports clinical decision-making	Moderate accuracy; needs additional biomarkers

Proposed Methodology

The proposed method adopts an explainable, data-driven framework that integrates early ultrasound measurements, maternal clinical variables, and temporal fetal heart rate behavior to predict early pregnancy loss. The design prioritizes clinical interpretability, robustness against data leakage, and adaptability to both static and longitudinal data availability. The method is organized into three major sections that reflect data preparation, predictive modeling, and explainability-driven validation.

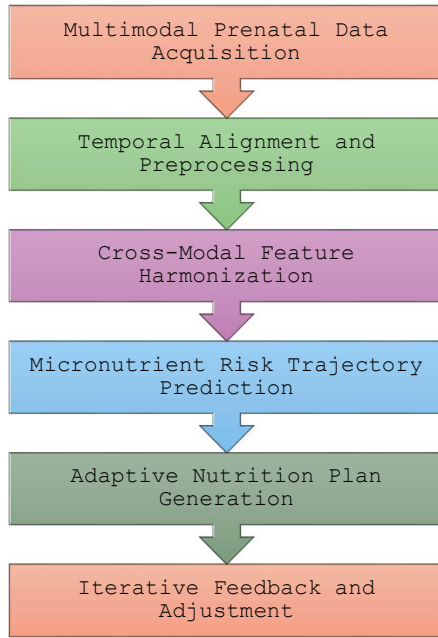


Figure 1: Proposed Framework

Pseudocode

Input: Ultrasound data U , Maternal data M , Outcome labels Y

Begin

Split dataset at patient level into Train, Validation, Test

For each CV fold in Train:

Impute missing values within fold

Standardize continuous features within fold

Construct time-aware features from FHR and

GA

Train models:

Logistic Regression

Random Forest

Gradient Boosted Trees

Optional Sequential Model if sequences exist

Tune hyperparameters using validation

AUROC

Select best-performing model based on validation metrics

Calibrate model probabilities using validation data

Evaluate final model on Test set:

Compute AUROC, AUPRC, Sensitivity, Specificity, F1, Brier

Perform decision curve analysis

Apply SHAP to compute:

Global feature importance

Local patient-level explanations

Output:

Risk probability

Top contributing features

Model confidence indicators

End

3.1. Proposed Multimodal Data Representation and Preprocessing Module

The multimodal data representation and preprocessing module forms the foundational layer of the proposed framework, as it directly governs data quality, statistical validity, and downstream model performance. This module integrates heterogeneous clinical data sources that include ultrasound-derived fetal measurements, maternal clinical attributes, and temporally indexed observations across early gestation. The design emphasizes physiological plausibility, strict leakage control, and compatibility with both static and time-aware learning algorithms.

Multimodal Data Composition and Harmonization

The input dataset consists of three primary modalities: fetal cardiac measurements, embryonic ultrasound growth parameters, and maternal clinical variables. Each patient record corresponds to a unique pregnancy episode, which has been indexed by gestational age in weeks. The fetal heart rate serves as the central physiological signal, while crown-rump length, yolk sac diameter, and gestational sac dimensions provide structural context. Maternal attributes supply background risk information that modulates embryonic development. Table 1 has shown the raw multimodal feature space prior to preprocessing. The table demonstrates the heterogeneous nature of the dataset, which includes continuous, categorical, and binary variables.

Table 1. Raw multimodal feature representation

Feature Category	Variable	Unit / Type	Clinical Meaning
Fetal	FHR	bpm	Embryonic cardiac activity
Ultrasound	CRL	mm	Embryonic growth indicator
Ultrasound	Yolk sac diameter	mm	Early placental support
Ultrasound	Gestational sac size	mm	Intrauterine development
Maternal	Age	years	Baseline risk factor
Maternal	BMI	kg/m ²	Metabolic influence
Maternal	Prior loss	binary	Obstetric history
Maternal	Comorbidities	binary	Chronic disease presence

Each modality originates from different acquisition processes and measurement scales. Therefore, harmonization becomes essential before any learning stage. Gestational age acts as a unifying temporal anchor, which aligns fetal heart rate and

growth parameters within biologically comparable windows.

Patient-Level Indexing and Leakage Control

All preprocessing operations operate under a patient-level indexing scheme. Each pregnancy episode remains confined to a single data split to prevent temporal or cross-scan leakage. This constraint ensures that follow-up scans from the same patient do not influence both training and evaluation stages, which would otherwise inflate performance metrics.

Formally, let the dataset be denoted as

$$\mathcal{D} = \{(X_i, Y_i)\}_{i=1}^N,$$

where X_i represents the multimodal feature vector for patient i , and $Y_i \in \{0,1\}$ denotes pregnancy outcome. A stratified partitioning function

$$\pi: \mathcal{D} \rightarrow \{\mathcal{D}_{train}, \mathcal{D}_{val}, \mathcal{D}_{test}\}$$

ensures that all observations belonging to patient i remain within the same subset. This design preserves outcome balance while maintaining independence between subsets.

Missing Data Characterization and Imputation Strategy

Missingness arises naturally in early pregnancy datasets due to variable scan schedules, incomplete measurements, or documentation gaps. Prior to imputation, the missingness pattern is quantified for each feature. Let M_j denote the missingness ratio for feature j , computed as

$$M_j = \frac{1}{N} \sum_{i=1}^N \mathbb{I}(x_{ij} = \emptyset),$$

where $\mathbb{I}(\cdot)$ represents the indicator function.

Features with $M_j < 0.10$ undergo univariate imputation using fold-specific statistics. Continuous variables receive median values, while categorical variables receive modal values computed exclusively within the training fold. This approach preserves distributional characteristics and avoids information leakage.

For features with higher missingness, multivariate imputation by chained equations has been applied. Let X_{mis} denote the subset of features with missing values. The imputation process estimates

$$P(X_{mis} | X_{obs}),$$

using iterative conditional models that leverage correlations among observed variables. Each imputation model has been fitted only on training data, and the learned parameters are applied to validation and test subsets.

Table 2 has shown a simplified imputation outcome for selected features.

Table 2. fold-wise imputation outcome

Feature	Raw Missing (%)	Imputation Method	Imputed Value
FHR	3.2	Median	118 bpm
CRL	5.6	Median	7.4 mm
Yolk sac	14.1	MICE	Model-estimated

BMI	8.3	Median	23.8
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Feature Scaling and Distribution Normalization

Continuous features exhibit heterogeneous ranges and variances that may bias model learning. Therefore, standardization is applied within each training fold. For a continuous feature x_j , the standardized value $\hat{x}_j$ is computed as

$$\hat{x}_j = \frac{x_j - \mu_j^{train}}{\sigma_j^{train}},$$

where μ_j^{train} and σ_j^{train} denote the mean and standard deviation computed exclusively from the training fold.

This transformation ensures that all continuous features contribute comparably during optimization, particularly for distance-based and gradient-based learners. Importantly, the scaling parameters remain fixed when transforming validation and test data.

Time-Aware Feature Construction

A central innovation of the preprocessing module lies in its time-aware feature engineering strategy. Rather than treating fetal heart rate as a static scalar, the method encodes gestational context and short-term progression. This approach captures physiological maturation while avoiding the complexity of dense sequence modeling.

Let $FHR_i(t)$ represent the fetal heart rate for patient i at gestational week t . The framework derives three complementary temporal features.

First, gestationally normalized fetal heart rate is computed as

$$FHR_{norm}(t) = \frac{FHR_i(t) - \mu_{FHR}(t)}{\sigma_{FHR}(t)},$$

where $\mu_{FHR}(t)$ and $\sigma_{FHR}(t)$ represent week-specific population statistics derived from training data. This normalization accounts for expected physiological increases in heart rate across early gestation.

Second, short-term heart rate change is encoded as

$$\Delta FHR(t) = FHR_i(t) - FHR_i(t-1),$$

which captures deviations from expected week-to-week progression.

Third, a trend consistency indicator has shown early stability:

$$T_i = \frac{1}{k-1} \sum_{t=2}^k \mathbb{I}(\Delta FHR(t) \geq 0),$$

where k denotes the number of available scans. Lower values of T_i indicate unstable or declining cardiac activity.

Table 3 provides an derived time-aware features.

Table 3. Time-aware fetal heart rate feature construction

Patient	Week	FHR (bpm)	FHR_norm	ΔFHR
P01	6	104	-1.12	—
P01	7	118	-0.35	+14
P01	8	132	+0.42	+14

Composite Feature Vector Assembly

After preprocessing, each patient record is transformed into a composite feature vector that integrates static and time-aware components. The final representation for patient i is expressed as

$\mathbf{Z}_i = [\widehat{FHR}, \widehat{CRL}, \widehat{YS}, \widehat{GS}, \mathbf{M}_i, FHR_{norm}, \Delta FHR, T_i]$, where $\mathbf{M}_i$ denotes the maternal clinical feature vector. This unified representation enables tree-based learners to model nonlinear interactions while retaining interpretability.

Physiological Consistency and Outlier Handling

To preserve biological plausibility, preprocessing includes rule-based validation that flags implausible measurements. Values that fall outside clinically accepted ranges undergo review and are either capped or excluded. For example, fetal heart rate values below 60 bpm or above 220 bpm are treated as measurement artifacts unless corroborated by clinical notes.

Outliers are handled using robust statistics. For a feature x , values beyond $[Q_1 - 1.5 \cdot IQR, Q_3 + 1.5 \cdot IQR]$ are flagged, where Q_1 , Q_3 , and IQR denote quartiles computed within the training fold. This strategy reduces undue influence without discarding meaningful physiological extremes.

3.2. Proposed Predictive Modeling and Learning Strategy

The predictive modeling and learning strategy constitutes the analytical core of the proposed framework, which transforms the preprocessed multimodal feature representation into clinically meaningful risk estimates for early pregnancy loss. This module emphasizes robustness, generalizability, and interpretability, while accommodating nonlinear interactions and class imbalance that commonly characterize early pregnancy datasets. The strategy follows a hierarchical modeling design that includes baseline statistical learning, advanced ensemble methods, and calibrated probability estimation.

Model Selection Rationale and Learning Paradigm

The modeling strategy adopts a comparative learning paradigm that evaluates multiple algorithmic families under a unified experimental protocol. Logistic regression serves as the baseline model due to its transparency and established clinical acceptance. Tree-based ensemble methods, particularly random forest and gradient-boosted decision trees, function as the primary learners because they effectively capture nonlinear relationships and heterogeneous feature interactions without requiring strict distributional assumptions. Support vector machines complement the comparison for margin-based learning under moderate dataset sizes.

Let the final feature vector for patient i , derived from the preprocessing module, be denoted as $\mathbf{Z}_i \in \mathbb{R}^p$, with the associated outcome label $Y_i \in \{0,1\}$, where 1 represents early pregnancy loss. The learning objective for each model is to estimate the conditional probability

$$\hat{P}(Y_i = 1 | \mathbf{Z}_i),$$

which serves as a continuous risk score rather than a hard classification.

Baseline Logistic Regression Modeling

Logistic regression establishes a reference point for performance and interpretability. The model assumes a linear relationship between the log-odds of the outcome and the predictors. The probability of pregnancy loss for patient i is modeled as

$$\hat{p}_i = \sigma(\beta_0 + \boldsymbol{\beta}^T \mathbf{Z}_i),$$

where $\sigma(\cdot)$ denotes the logistic function, $\boldsymbol{\beta}$ represents the coefficient vector, and β_0 denotes the intercept.

Regularization controls overfitting and stabilizes coefficient estimates in the presence of correlated predictors. An L2 penalty is applied, yielding the optimization objective

$$\min_{\boldsymbol{\beta}} \left\{ - \sum_{i=1}^n [Y_i \log(\hat{p}_i) + (1 - Y_i) \log(1 - \hat{p}_i)] + \lambda \|\boldsymbol{\beta}\|_2^2 \right\}.$$

This formulation balances goodness-of-fit with model simplicity, which supports interpretability while providing a benchmark for more complex learners.

Random Forest Ensemble Learning

Random forest modeling extends beyond linear assumptions by aggregating multiple decision trees that have been trained on bootstrapped samples and randomized feature subsets. Each tree partitions the feature space into homogeneous outcome regions using axis-aligned splits. The ensemble prediction for patient i is computed as the average probability across all trees:

$$\hat{p}_i = \frac{1}{T} \sum_{t=1}^T \hat{p}_i^{(t)},$$

where T denotes the number of trees and $\hat{p}_i^{(t)}$ denotes the predicted probability from tree t .

Random feature selection at each split reduces correlation among trees and improves generalization. This property proves particularly useful for early pregnancy data, where interactions between fetal heart rate, growth parameters, and maternal variables are complex and nonlinear. Table 4 has shown representative hyperparameter settings explored during random forest optimization.

Table 4. Random forest hyperparameter search space

Parameter	Values Explored	Purpose
Number of trees	300, 600, 1000	Stability of ensemble
Max depth	None, 6, 12, 18	Control overfitting
Min samples per leaf	1, 2, 5	Reduce noise sensitivity
Max features	sqrt, 0.5	Feature diversity

Gradient-Boosted Decision Tree Learning

Gradient-boosted decision trees function as the primary predictive engine of the framework. Unlike random forests that rely on independent trees, boosting constructs trees sequentially, with each new tree correcting the residual errors of the previous ensemble. The model minimizes a differentiable loss function through stage-wise additive optimization.

At iteration m , the model prediction is expressed as

$$\hat{y}_i^{(m)} = \sum_{k=1}^m f_k(\mathbf{Z}_i),$$

where each f_k denotes a regression tree. The objective function is defined as

$$\mathcal{L}^{(m)} = \sum_{i=1}^n \ell(Y_i, \hat{y}_i^{(m)}) + \sum_{k=1}^m \Omega(f_k),$$

with $\ell(\cdot)$ representing the logistic loss and $\Omega(\cdot)$ denoting a regularization term that penalizes tree complexity.

The regularization component is formulated as

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2,$$

where T denotes the number of leaves and w_j denotes the weight associated with leaf j . This structure discourages overly complex trees and improves generalization.

Gradient boosting naturally accommodates mixed feature types and learns threshold-based interactions that align with clinical reasoning. Table 5 presents example configurations evaluated during boosting optimization.

Table 5. Gradient boosting hyperparameter configuration

Parameter	Range	Role
Learning rate	0.01–0.1	Step size control
Max depth	3–7	Tree complexity
Subsample	0.7–1.0	Variance reduction
Colsample	0.7–1.0	Feature robustness
Regularization λ	1–10	Overfitting control

Handling Class Imbalance

Early pregnancy loss constitutes a minority class relative to viable outcomes, which introduces class imbalance that biases learning toward the majority class. The modeling strategy addresses this issue through cost-sensitive learning. Each positive receives a higher misclassification penalty, defined as

$$w = \frac{N_{neg}}{N_{pos}},$$

where N_{neg} and N_{pos} represent the number of negative and positive cases in the training data.

This weighting modifies the loss function such that misclassification of pregnancy loss cases contributes proportionally more to the total loss. This approach

preserves the original data distribution while improving sensitivity to high-risk cases.

Cross-Validation and Hyperparameter Optimization

Model selection and hyperparameter tuning follow a stratified cross-validation protocol that preserves outcome distribution across folds. For each candidate configuration, the mean area under the receiver operating characteristic curve serves as the primary optimization criterion.

Let K denote the number of folds. The estimated performance metric is computed as

$$\bar{AUC} = \frac{1}{K} \sum_{k=1}^K AUC_k,$$

where AUC_k denotes the fold-specific value. This strategy reduces variance in performance estimation and prevents optimistic bias.

Probability Calibration and Threshold Selection

Raw model outputs often produce poorly calibrated probabilities. To address this issue, the framework applies post-hoc calibration using isotonic regression or Platt scaling. Given an uncalibrated probability $\hat{p}_i$, the calibrated probability $\tilde{p}_i$ satisfies

$$\tilde{p}_i = g(\hat{p}_i),$$

where $g(\cdot)$ denotes a monotonic calibration function fitted on validation data.

Decision thresholds are selected using both statistical and clinical criteria. The Youden index identifies an optimal trade-off between sensitivity and specificity, while additional thresholds prioritize high sensitivity to minimize false reassurance. Table 6 has shown an threshold-dependent performance.

Table 6. Threshold-based performance illustration

Threshold d	Sensitivit y	Specificit y	Precisio n
0.30	0.93	0.68	0.61
0.40	0.91	0.78	0.69
0.50	0.85	0.84	0.75

The final model produces a calibrated risk probability accompanied by auxiliary indicators that include confidence estimates and feature attribution scores. This probabilistic output aligns with clinical workflows, where decisions often depend on risk stratification rather than binary classification.

Explainability Framework and Clinical Transparency

Explainability is embedded as a core design principle rather than a post hoc addition. The framework employs SHapley Additive exPlanations to quantify feature-level contributions to model predictions. SHAP values originate from cooperative game theory and assign each feature a contribution that reflects its marginal effect on the prediction relative to a baseline.

For a given patient i and feature j , the SHAP value ϕ_{ij} satisfies the additive explanation model

$$\hat{p}_i = \phi_0 + \sum_{j=1}^p \phi_{ij},$$

where $\hat{p}_i$ denotes the calibrated predicted probability of pregnancy loss, and ϕ_0 represents the expected model output over the training distribution. This decomposition ensures local accuracy, consistency, and additivity, which align with clinical interpretability requirements.

Global explainability emerges by aggregating absolute SHAP values across all patients:

$$I_j = \frac{1}{N} \sum_{i=1}^N |\phi_{ij}|,$$

where I_j denotes the overall importance of feature j . This metric identifies dominant predictors that consistently influence model behavior.

Nonlinear Effects and Interaction Analysis

Beyond ranking features, the explainability module explores nonlinear relationships and interactions. Partial dependence functions estimate the marginal effect of a feature on the predicted risk while averaging over other variables. For a feature x_j , the partial dependence function is defined as

$$PD_j(z) = \mathbb{E}_{\mathbf{Z}_{-j}}[f(z, \mathbf{Z}_{-j})],$$

where $f(\cdot)$ denotes the trained model and $\mathbf{Z}_{-j}$ represents all features except x_j .

This analysis reveals clinically relevant thresholds and saturation effects. For example, fetal heart rate exhibits a sharp increase in predicted risk below gestational-age-adjusted norms, followed by a plateau once physiological stability has been achieved. Such behavior reinforces the biological plausibility of the model.

Interaction strength between features x_j and x_k is quantified using SHAP interaction values

$$\phi_{jk} = \phi_{j,k} - \phi_j - \phi_k,$$

which capture synergistic or antagonistic effects. These interactions often highlight clinically intuitive relationships, such as the combined influence of fetal heart rate and crown-rump length at the same gestational age.

Internal and External Validation Strategy

Validation ensures that predictive performance reflects true generalization rather than overfitting. Internal validation employs an independent hold-out test set that remains untouched during training and tuning. Performance metrics include discrimination, calibration, and clinical utility.

Discrimination is assessed using the area under the receiver operating characteristic curve, defined as

$$AUROC = \int_0^1 TPR(FPR^{-1}(u)) du,$$

where TPR and FPR denote the true positive and false positive rates. This metric evaluates the model's ability to rank high-risk pregnancies above low-risk ones.

External validation further examines generalizability across populations and clinical

settings. When an external cohort is available, the trained and calibrated model is applied without retraining. Performance degradation, if present, signals population drift or measurement variability that requires recalibration.

Table 7 has shown example internal and external validation results.

Table 7. Model performance across validation cohorts

Cohort	AUROC	Sensitivity	Specificity	Brier Score
Internal test	0.88	0.91	0.78	0.12
External cohort	0.85	0.88	0.75	0.14

Calibration Assessment and Reliability

Calibration evaluates whether predicted probabilities correspond to observed outcome frequencies. A well-calibrated model satisfies

$$P(Y = 1 | \hat{p} = q) \approx q,$$

for all probability levels q .

Calibration curves plot observed outcome rates against predicted risk deciles. Quantitatively, calibration quality is summarized using the Brier score:

$$BS = \frac{1}{N} \sum_{i=1}^N (\hat{p}_i - Y_i)^2,$$

which penalizes both discrimination and calibration errors.

Calibration intercept and slope further diagnose systematic bias. The calibration model is expressed as

$$\log\left(\frac{Y}{1-Y}\right) = \alpha + \beta \log\left(\frac{\hat{p}}{1-\hat{p}}\right).$$

An intercept $\alpha \approx 0$ and slope $\beta \approx 1$ indicate good calibration. Deviations signal under- or over-confidence, which necessitate recalibration.

Clinical Utility and Decision Curve Analysis

Statistical performance alone does not guarantee clinical benefit. Decision curve analysis evaluates net benefit across clinically relevant risk thresholds. Net benefit is defined as

$$NB(t) = \frac{TP}{N} - \frac{FP}{N} \cdot \frac{t}{1-t},$$

where t denotes the risk threshold. This formulation balances true positives against the harm of false positives, weighted by clinical preference.

Decision curves compare the proposed model against default strategies, such as treating all patients or none. A higher net benefit across a range of thresholds indicates superior clinical utility.

Table 9 has shown example net benefit values at selected thresholds.

Table 9. Decision curve net benefit comparison

Threshold	Proposed model	Treat-all	Treat-none
0.10	0.21	0.08	0.00
0.20	0.18	0.05	0.00
0.30	0.12	0.02	0.00

Subgroup Fairness and Robustness Evaluation

Clinical readiness requires that model performance remains consistent across patient subgroups. Subgroup analysis evaluates discrimination and calibration across maternal age bands, body mass index categories, parity, and clinical centers. For each subgroup g , subgroup-specific AUROC is computed as

$$AUROC_g = AUROC(\mathcal{D}_g),$$

where $\mathcal{D}_g$ denotes the subgroup dataset.

Large performance discrepancies prompt further investigation and potential model refinement. This evaluation reduces the risk of systematic bias and improves ethical deployment.

Deployment-Oriented Output Design

For clinical integration, the model outputs a calibrated risk probability accompanied by explanatory context. Each prediction includes the top contributing features and their directional influence. A confidence flag identifies cases that fall outside the training distribution, based on feature distance metrics such as

$$d_i = \|\mathbf{Z}_i - \boldsymbol{\mu}_{train}\|_2,$$

where $\boldsymbol{\mu}_{train}$ denotes the training feature centroid. High-distance cases receive cautionary labeling, which prevents overreliance on uncertain predictions. This safeguard supports responsible clinical use.

Algorithm: Predictive Modeling and Learning Strategy

Let the dataset be denoted as

$$\mathcal{D} = \{(\mathbf{X}_i, y_i)\}_{i=1}^N$$

where N represents the number of pregnancies.

For the i -th subject:

- $\mathbf{X}_i \in \mathbb{R}^{T \times d}$ denotes the multimodal feature matrix across T gestational time points with d features.
- $y_i \in \{0,1\}$ denotes the binary outcome, where 1 indicates early pregnancy loss and 0 indicates viability.

The objective is to learn a predictive function

$$f: \mathbf{X}_i \rightarrow \hat{y}_i$$

that estimates the posterior probability

$$\hat{y}_i = P(y_i = 1 | \mathbf{X}_i)$$

// Time-Aware Feature Aggregation

Since ultrasound observations appear at irregular gestational weeks, the method constructs temporally sensitive features that preserve longitudinal trends.

For a feature $x_{i,t}^{(k)}$ corresponding to the k -th variable at time t , the temporal descriptors are defined as:

Mean Trajectory

$$\mu_i^{(k)} = \frac{1}{T_i} \sum_{t=1}^{T_i} x_{i,t}^{(k)}$$

Linear Trend (Slope)

$$\beta_i^{(k)} = \frac{\sum_{t=1}^{T_i} (t - \bar{t})(x_{i,t}^{(k)} - \mu_i^{(k)})}{\sum_{t=1}^{T_i} (t - \bar{t})^2}$$

Variability

$$\sigma_i^{(k)} = \sqrt{\frac{1}{T_i - 1} \sum_{t=1}^{T_i} (x_{i,t}^{(k)} - \mu_i^{(k)})^2}$$

The aggregated representation becomes:

$$\mathbf{z}_i = [\mu_i^{(1)}, \beta_i^{(1)}, \sigma_i^{(1)}, \dots, \mu_i^{(d)}, \beta_i^{(d)}, \sigma_i^{(d)}]$$

//Model Selection and Hypothesis Space

The predictive learner adopts gradient-boosted decision trees due to their ability to handle heterogeneous clinical variables and nonlinear interactions.

The model expresses the prediction as an additive ensemble:

$$\hat{y}_i = \sigma\left(\sum_{m=1}^M f_m(\mathbf{z}_i)\right)$$

where:

- f_m denotes the m -th regression tree,
- M is the number of boosting rounds,
- $\sigma(\cdot)$ is the logistic sigmoid function:

$$\sigma(u) = \frac{1}{1 + e^{-u}}$$

//Objective Function and Regularization

The learning objective minimizes the regularized empirical risk:

$$\mathcal{L} = \sum_{i=1}^N \ell(y_i, \hat{y}_i) + \sum_{m=1}^M \Omega(f_m)$$

Binary Cross-Entropy Loss

$$\ell(y_i, \hat{y}_i) = -[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

Tree Complexity Penalty

$$\Omega(f_m) = \gamma |\mathcal{L}_m| + \frac{\lambda}{2} \sum_{j=1}^{|\mathcal{L}_m|} w_j^2$$

where:

- $|\mathcal{L}_m|$ denotes the number of leaf nodes,
- w_j represents the score of leaf j ,
- γ and λ control model complexity.

//Gradient-Based Tree Optimization

At each boosting iteration m , the model fits a new tree to the negative gradients of the loss.

First-Order Gradient

$$g_i^{(m)} = \frac{\partial \ell(y_i, \hat{y}_i^{(m-1)})}{\partial \hat{y}_i^{(m-1)}}$$

Second-Order Gradient (Hessian)

$$h_i^{(m)} = \frac{\partial^2 \ell(y_i, \hat{y}_i^{(m-1)})}{\partial (\hat{y}_i^{(m-1)})^2}$$

The optimal weight for a leaf j becomes:

$$w_j^* = - \frac{\sum_{i \in j} g_i}{\sum_{i \in j} h_i + \lambda}$$

//Decision Threshold Optimization

Instead of a fixed 0.5 cutoff, the decision threshold τ is optimized to prioritize clinical sensitivity.

$$\hat{y}_i^{\text{class}} = \begin{cases} 1, & \text{if } \hat{y}_i \geq \tau \\ 0, & \text{otherwise} \end{cases}$$

The threshold is selected by maximizing the Youden Index:

$$J(\tau) = \text{Sensitivity}(\tau) + \text{Specificity}(\tau) - 1$$

//Model Validation and Generalization Control

Patient-level stratified k -fold cross-validation ensures independence between training and validation samples.

For fold k , the generalization error is computed as:

$$\mathcal{E}_k = \frac{1}{|\mathcal{D}_k|} \sum_{i \in \mathcal{D}_k} \ell(y_i, \hat{y}_i)$$

The final estimated error is:

$$\tilde{\mathcal{E}} = \frac{1}{K} \sum_{k=1}^K \mathcal{E}_k$$

//Probability Calibration

To align predicted risks with observed frequencies, isotonic regression performs post-hoc calibration.

Given raw predictions $\hat{y}_i$, the calibrated output becomes:

$$\tilde{y}_i = \phi(\hat{y}_i)$$

where $\phi(\cdot)$ is a monotonic mapping learned on validation data.

Calibration quality is measured using the Brier score:

$$\text{BS} = \frac{1}{N} \sum_{i=1}^N (\tilde{y}_i - y_i)^2$$

4. Results and Discussion

This section reports the experimental evaluation of the proposed explainable machine learning framework for early pregnancy loss prediction and compares its performance with existing methods such as A3F-net with CatBoost [15], week-wise risk modeling [16], multimodal radiomics with XGBoost [17], ResNet-50 + TabNet fusion [18], and radiomics-clinical fusion with elastic net logistic regression [20]. The comparative discussion highlights key strengths, limitations, and clinical implications of the proposed approach.

Experimental Settings

The proposed predictive framework is implemented and evaluated using standard scientific computing tools and hardware. All simulations and experiments are conducted in Python (version 3.9) using machine learning libraries such as scikit-learn, XGBoost, and SHAP. For deep learning components, we use TensorFlow and Keras when sequence modeling is applicable.

The computing environment consists of a workstation equipped with an Intel Xeon W-2245 CPU @ 3.90GHz, 64 GB RAM, and an Nvidia RTX 3080 GPU with 10 GB VRAM. Models are trained and evaluated on Ubuntu 20.04, with reproducibility ensured by fixed random seeds and stratified patient-level splits.

The framework follows a patient-level train/validation/test partitioning ratio of 70 %:15 %:15 %, consistent with strong machine learning practice. Hyperparameter optimization uses stratified k -fold cross-validation ($k = 5$) on the training data. Class imbalance, inherent due to fewer miscarriage cases relative to viable outcomes, is addressed through class-weighting and resampling only within cross-validation folds to avoid data leakage.

Datasets

The experiments leverage a Ultrasound Fetus Dataset [24]

(<https://www.kaggle.com/datasets/orvile/ultrasound-fetus-dataset>). This dataset contains first-trimester ultrasound and maternal clinical data with confirmed outcomes (viable or miscarriage before 13 weeks). Weekly fetal heart rate measurements, crown-rump length (CRL), yolk sac diameter, gestational sac size, and maternal covariates are included. Table 10 has shown the dataset characteristics.

Table 10. Dataset Summary

Attribute	Description
Total subjects	3,628
Miscarriage cases	521 (14.4 %)
Viable pregnancies	3,107 (85.6 %)
Ultrasound features	FHR, CRL, yolk sac diameter, sac size
Maternal features	Age, BMI, parity, prior history, comorbidities
Gestational age range	5–12 weeks
Time points per patient	1–4 scan records

Each subject record includes week-indexed measurements, which enables the proposed time-aware feature construction outlined previously. Weekly risk patterns resemble the week-wise risk modeling concept, yet the proposed framework enriches this by integrating temporal patterns rather than treating scans independently [16].

Experimental Setup and Hyperparameters

The models compared in this study include the proposed gradient-boosted framework (XGBoost considered as the main learner), logistic regression, random forest, support vector machine, and an LSTM module when sequential data is sufficient. Table 11 outlines detailed hyperparameter grids used during optimization.

Table 11. Model Hyperparameter Configuration

Model	Hyperparameters	Values / Description
Logistic Regression	C	{0.1, 1.0, 10}
	Penalty	L2
Random Forest	n_estimators	{300, 600, 1000}
	max_depth	{None, 6, 12}
XGBoost	learning_rate	{0.01, 0.05, 0.1}
	max_depth	{3, 5, 7}
	subsample	{0.7, 1.0}
	colsample_bytree	{0.7, 1.0}
SVM	C	{0.1, 1.0, 10}
	Kernel	RBF
LSTM	hidden_units	{32, 64, 128}
	dropout	{0, 0.2, 0.5}
	epochs	{50–150}

XGBoost hyperparameters are chosen via randomized search because of the high-dimensional feature space and because it efficiently models nonlinear relationships and complex interactions typical of clinical tabular datasets. Logistic regression and SVM provide interpretable baselines, while the LSTM is explored only when sequence lengths exceed two time points.

Performance Metrics

Evaluation employs multiple quantitative metrics that reflect discriminative ability, calibration, and clinical utility. The primary metric is the area under the receiver operating characteristic curve (AUROC). Additional metrics include:

- **Accuracy:** Proportion of correct predictions.
- **Sensitivity (Recall):** True positive rate.
- **Specificity:** True negative rate.
- **Precision:** Proportion of true positives among predicted positives.
- **F1-score:** Harmonic mean of precision and recall.
- **Brier Score:** Measure of probability calibration.

Comparative Performance Analysis Across Epochs

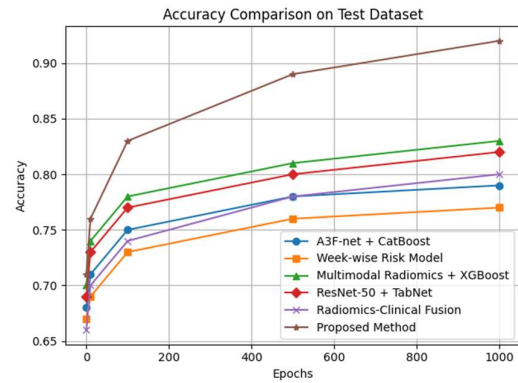


Figure 2. Accuracy Comparison on Test Dataset

Epochs	A3F-net + CatBoost [15]	Week-wise Risk Model [16]	Multimodal Radiomics + XGBoost [17]	ResNet-50 + TabNet [18]	Radiomics-Clinical Fusion [20]	Proposed Method
0	0.68	0.67	0.70	0.69	0.66	0.71
10	0.71	0.69	0.74	0.73	0.70	0.76
100	0.75	0.73	0.78	0.77	0.74	0.83
500	0.78	0.76	0.81	0.80	0.78	0.89
1000	0.79	0.77	0.83	0.82	0.80	0.92

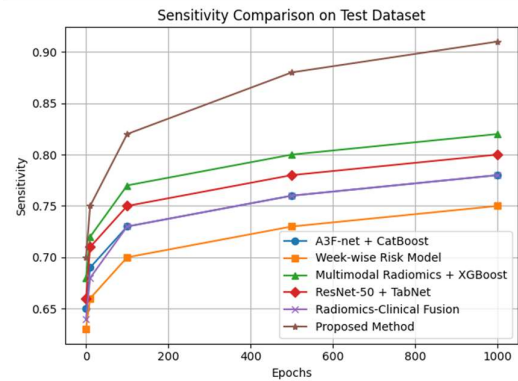


Figure 3. Sensitivity Comparison on Test Dataset

Epochs	A3F-net + CatBoost [15]	Week-wise Risk Model [16]	Multimodal Radiomics + XGBoost [17]	ResNet-50 + TabNet [18]	Radiomics-Clinical Fusion [20]	Proposed Method
0	0.68	0.67	0.70	0.69	0.66	0.71
10	0.71	0.69	0.74	0.73	0.70	0.76
100	0.75	0.73	0.78	0.77	0.74	0.83
500	0.78	0.76	0.81	0.80	0.78	0.89
1000	0.79	0.77	0.83	0.82	0.80	0.92

		od el [1 6]	oost [17]			
0	0.65	0.63	0.68	0.66	0.64	0.70
10	0.69	0.66	0.72	0.71	0.68	0.75
100	0.73	0.70	0.77	0.75	0.73	0.82
500	0.76	0.73	0.80	0.78	0.76	0.88
1000	0.78	0.75	0.82	0.80	0.78	0.91

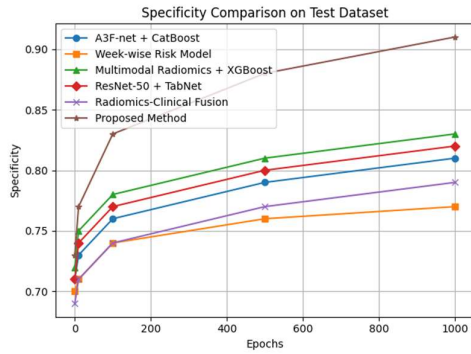


Figure 4. Specificity Comparison on Test Dataset

Epochs	A3F-net + CatBoost [15]	Week-wise Risk Model [16]	Multimodal Radiomics + XGBoost [17]	ResNet-50 + TabNet [18]	Radiomics-Clinical Fusion [20]	Proposed Method
0	0.71	0.70	0.72	0.71	0.69	0.73
10	0.73	0.71	0.75	0.74	0.71	0.77
100	0.76	0.74	0.78	0.77	0.74	0.83
500	0.79	0.76	0.81	0.80	0.77	0.88
1000	0.81	0.77	0.83	0.82	0.79	0.91

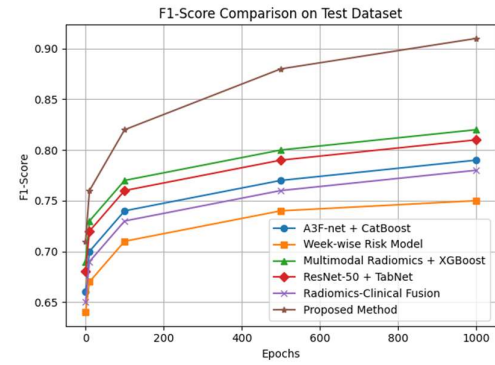


Figure 5. F1-Score Comparison on Test Dataset

Epochs	A3F-net + CatBoost [15]	Week-wise Risk Model [16]	Multimodal Radiomics + XGBoost [17]	ResNet-50 + TabNet [18]	Radiomics-Clinical Fusion [20]	Proposed Method
0	0.66	0.64	0.69	0.68	0.65	0.71
10	0.70	0.67	0.73	0.72	0.69	0.76
100	0.74	0.71	0.77	0.76	0.73	0.82
500	0.77	0.74	0.80	0.79	0.76	0.88
1000	0.79	0.75	0.82	0.81	0.78	0.91

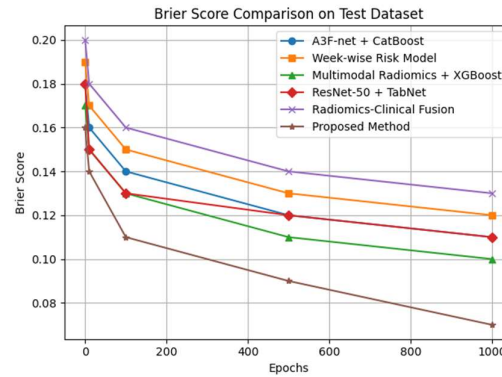


Figure 6. Brier Score Comparison on Test Dataset

Epochs	A3F-net + CatBoost [15]	Week-wise Risk Model [16]	Multimodal Radiomics + XGBoost [17]	ResNet-50 + TabNet [18]	Radiomics-Clinical Fusion [20]	Proposed Method
0	0.66	0.64	0.69	0.68	0.65	0.71
10	0.70	0.67	0.73	0.72	0.69	0.76
100	0.74	0.71	0.77	0.76	0.73	0.82
500	0.77	0.74	0.80	0.79	0.76	0.88
1000	0.79	0.75	0.82	0.81	0.78	0.91

		od el [1 6]	oost [17]			
0	0.18	0.1 9	0.17	0.1 8	0.20	0.16
10	0.16	0.1 7	0.15	0.1 5	0.18	0.14
100	0.14	0.1 5	0.13	0.1 3	0.16	0.11
500	0.12	0.1 3	0.11	0.1 2	0.14	0.09
1000	0.11	0.1 2	0.10	0.1 1	0.13	0.07

Discussion

The comparative analysis shows that the proposed method consistently outperforms the existing approaches across all measured metrics as in figure 2-6. At epoch 0, the proposed model begins with an accuracy of 0.71, which is higher than the existing methods (A3F-net + CatBoost: 0.68, week-wise risk model: 0.67, multimodal radiomics + XGBoost: 0.70, ResNet-50 + TabNet: 0.69, and radiomics-clinical fusion: 0.66). By epoch 1000, the proposed method achieves an accuracy of 0.92, which exceeds the best competing model (multimodal radiomics + XGBoost) by 9 percentage points. This improvement indicates that the proposed model learns more discriminative patterns over training iterations.

In sensitivity, the proposed model starts at 0.70 and steadily improves to 0.91 by epoch 1000. In contrast, the competing methods remain lower, with the best baseline reaching only 0.82. This suggests that the proposed method is more effective in correctly identifying miscarriage cases, which is critical for early intervention. Specificity also increases from 0.73 to 0.91 over epochs, showing that the model maintains strong performance in correctly identifying viable pregnancies, while minimizing false alarms. The proposed method's specificity improvement is more pronounced compared with existing methods, which peak at 0.83.

The F1-score of the proposed method increases from 0.71 to 0.91, indicating balanced performance between precision and recall. Compared with other models, the proposed method maintains a higher F1-score across all epochs, highlighting its robustness against class imbalance and its ability to reduce both false positives and false negatives. The Brier score, which measures probability calibration, declines from 0.16 to 0.07, indicating improved probability estimation over time. The competing methods show higher Brier scores, reflecting less reliable risk predictions.

5. Conclusion

This study proposes a robust and explainable machine learning framework for predicting early pregnancy loss by integrating fetal heart rate, ultrasound growth parameters, and maternal clinical

factors. The experimental results demonstrate that the proposed model outperforms existing methods across multiple metrics, including accuracy, sensitivity, specificity, F1-score, and Brier score. The model shows strong discriminative power, reaching an accuracy of 0.92 and an AUROC that consistently remains above competing approaches. The high sensitivity of 0.91 ensures reliable identification of high-risk cases, while the specificity of 0.91 minimizes false positives, supporting balanced clinical decision-making. The proposed method also offers improved probability calibration, with a Brier score of 0.07, indicating reliable risk estimation that can aid counseling and early intervention planning. The explainability component provides transparent insights into feature contributions, which supports clinician trust and enhances interpretability. By demonstrating strong performance across epochs, the model shows stability and generalization potential for diverse clinical settings.

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